

PRINAN SIL

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PROFESSIONAL SUMMARY

Data Engineer with 1.7+ years at Tata Consultancy Services, building and optimizing Oracle PL/SQL pipelines, ETL batch workflows, and API backends for enterprise insurance systems processing 10M+ records. Experienced in stored procedures, dynamic SQL, batch processing, query optimization, and transaction management at scale. Academic background in ML and bioinformatics (ACM India Award 2024). GATE AIR 488 in Data Analytics & AI. Targeting Data Engineer, ETL Developer, and Backend Data Engineer roles across GCCs, product companies, and MNCs.

PROFESSIONAL SKILLS

Programming: Java, Python, SQL, PL/SQL, JavaScript, C

Backend Technologies: Spring Boot, REST APIs, Microservices, JPA/Hibernate, Postman, Swagger

Databases: Oracle SQL, PL/SQL, PostgreSQL, MySQL

Tools / Platforms: Databricks, AWS, GCP, Azure, Git, GitHub, GitHub Copilot, Jenkins, CI/CD, VS Code

Core Skills: Stored Procedures, Cursors, Dynamic SQL, Views, Triggers, Batch Processing, Query Optimization, ETL Pipelines, Transaction Management, Performance Tuning, Schema Design, RCA Analysis

EXPERIENCE

Tata Consultancy Services (TCS)

Systems Engineer (Nov 2024 - Present)

Data Engineering

- Developed PL/SQL batch processing ETL pipelines for auto-rejecting deficient claims with 100% accuracy. Reducing processing time from 10 mins to 2mins.
- Automated reporting and validation ensuring high data accuracy and operational consistency across 10M+ records.
- Implemented claim deductible logic to automate financial adjustments across millions of records with high data integrity.
- Built a TDS procedure that processes over 10,000+ monthly transactions valued between 20 - 30 lakhs with zero errors.
- Diagnosed and resolved 500+ system defects through log analysis, debugging, root cause analysis, and performance tuning.

Pipeline & API Development

- Built REST APIs using Spring Boot to automate claim creation, reducing manual effort and processing time by 90%. Execution time reduced from 2 mins to 10 seconds.
- Delivered Short Payment feature, enabling automated secondary claims with 100% reconciliation accuracy.
- Developed a bulk approval module using optimized SQL, increasing throughput by 50% and reducing bottlenecks.
- Led the end-to-end rollout of the claims product, including schema design, rules, validation, and deployment.
- Reduced production issues by 70% through collaboration with business, QA and support teams on structural fixes.
- Integrated third-party API services for fraud detection and AI client screening, improving data governance and reliability.

Amazon ML Summer School

Student Apprentice (Sep 2023 - Nov 2023)

- Selected among 40,000+ applicants for Amazon's ML program focused on scalable ML systems and AI engineering.
- Solved ML problems using supervised, unsupervised, and deep learning techniques on real-world datasets.
- Analyzed scalable ML architectures and production level optimization strategies under mentorship of Amazon scientists.

PROJECTS / RESEARCH

Biomarker Discovery for Breast Cancer Subtype Detection *Python, Pandas, TensorFlow, Keras, Sklearn, NumPy*

- Built ML and DL models on genomic datasets for biomarker identification, achieving 90.74% classification accuracy.
- Processed large-scale genetic data using cleaning, normalization, and feature engineering, improving performance by 30%.
- Performed hyperparameter tuning and model evaluation using F1-Score and AUC-ROC metrics for accurate prediction.
- Generated data-driven insights supporting personalized treatment strategies and predictive healthcare analytics.
- Awarded ACM India Dissertation Award 2024 for research excellence in Machine Learning and Bioinformatics.

Projects on Data Analysis and Data Visualization

Tableau, Power BI, MS Excel

- Electric Vehicle Sales Analysis Using Tableau | **Live Demo**
- COVID-19 Case Analysis Using Power BI | **Github Link**

EDUCATION

Techno Main Salt Lake (B.Tech in Computer Science Engineering)

Kolkata, India

CGPA: 9.24

2020 - 2024

CERTIFICATIONS

- GH-300: GitHub Copilot Fundamentals - **Microsoft Solutions Partner**
- Databricks Fundamentals Accreditation - **Databricks Academy**
- Computation using Python by IIT Madras - **NPTEL**

HONORS & AWARDS

ACM India East and North East Bachelor Dissertation Award 2024

May 2024

- Led a 4 member team for ML based bioinformatics research with strong collaboration and problem solving.
- Awarded 3rd Prize among 36 finalists from over 200 engineering colleges.

ACHIEVEMENTS

- Secured AIR 488 in GATE 2024 (Data Analytics & Artificial Intelligence).
- Successfully completed the Google Cloud Ready Facilitator Program 2022.
- Achieved Star of the month award for Project Excellence in Rapid Issue analysis, resolution and closure.